CardamomOT — analysis report

`/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Semrau`

Generated on 2026-10-25 17:29

Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	41
Stimuli	1
Observed cells	1714
Timepoints	0, 6, 12, 24, 36, 48, 60, 72, 96
Cell types	Pluripotency, Post-implantation_Epiblast, Extraembryonic_Endoderm, Neuroectoderm
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic)

Perturbations (perturbation_simulation)

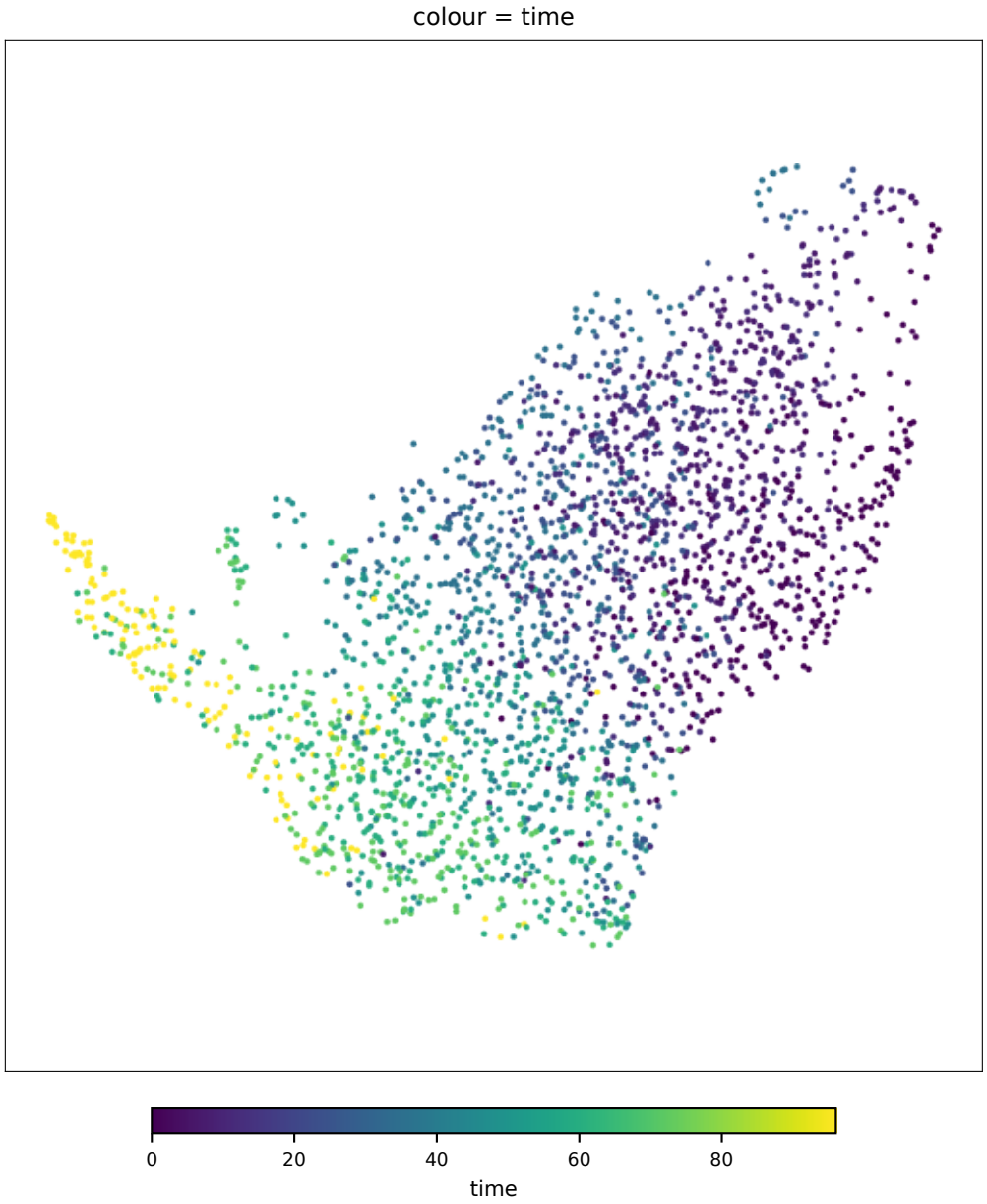
- ✓ P1 KO Dnmt3a
- ✓ P2 OV Dnmt3a

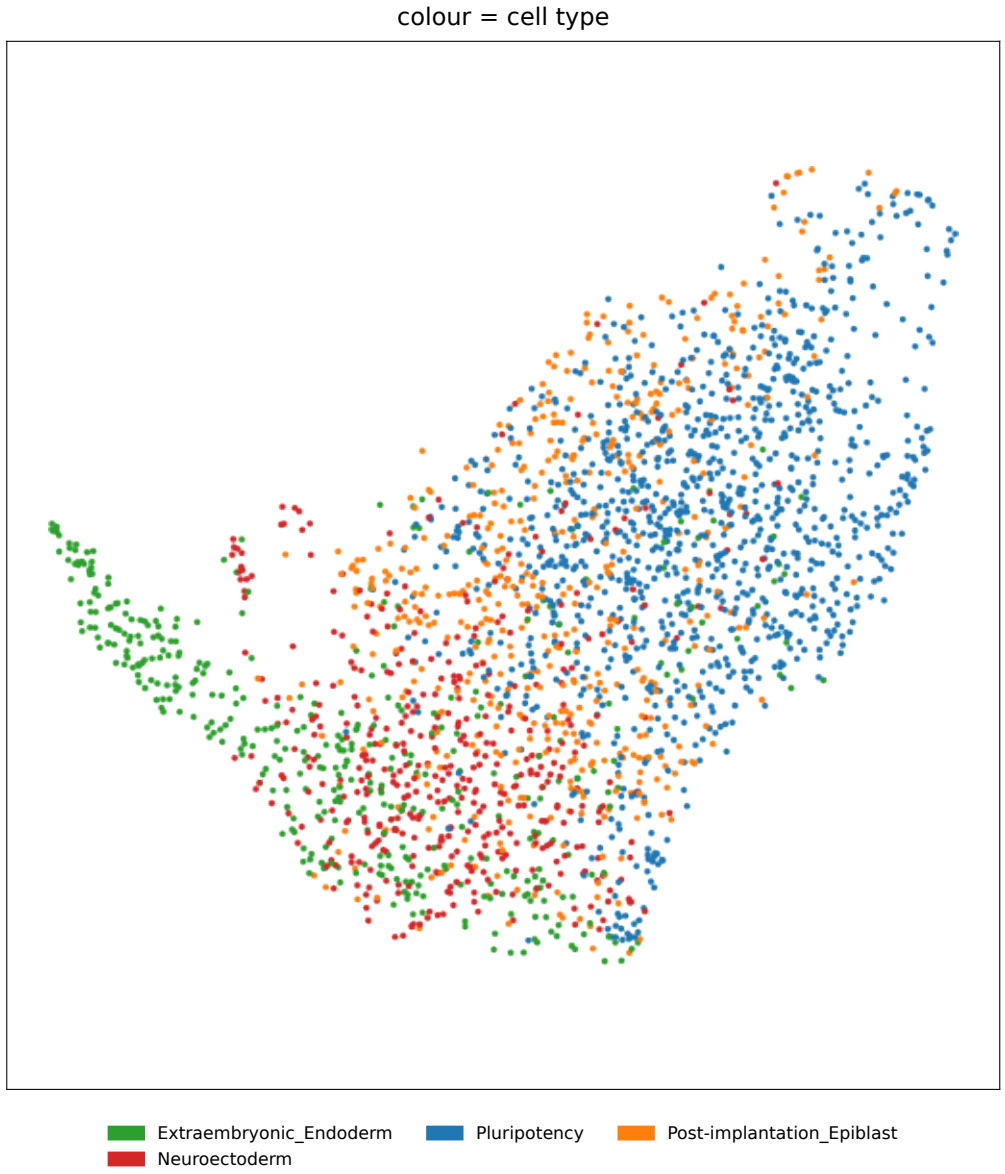
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

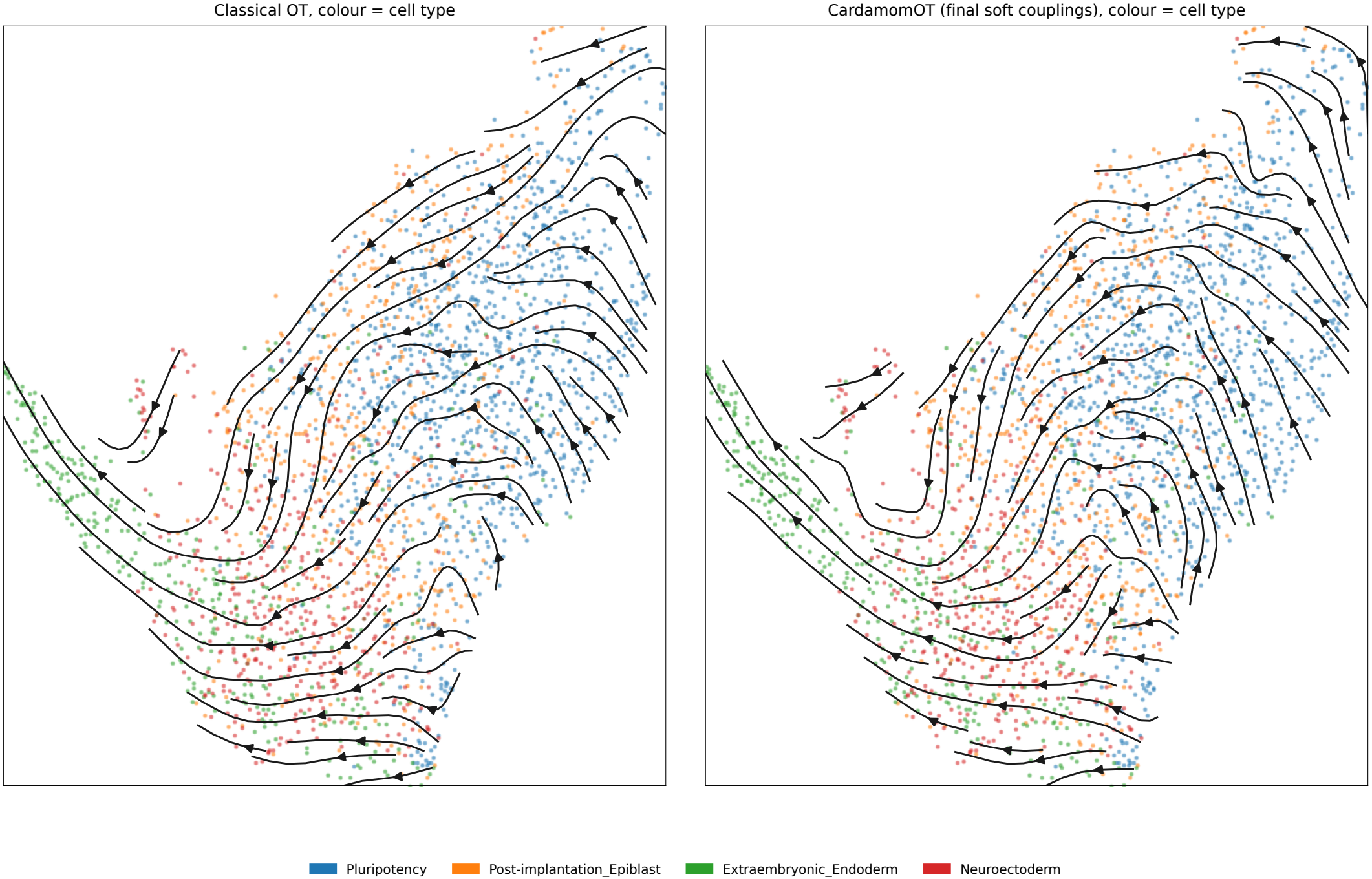
Data/data.h5ad, 2449 cells: log1p(counts), 41 highly variable genes, PCA (40 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





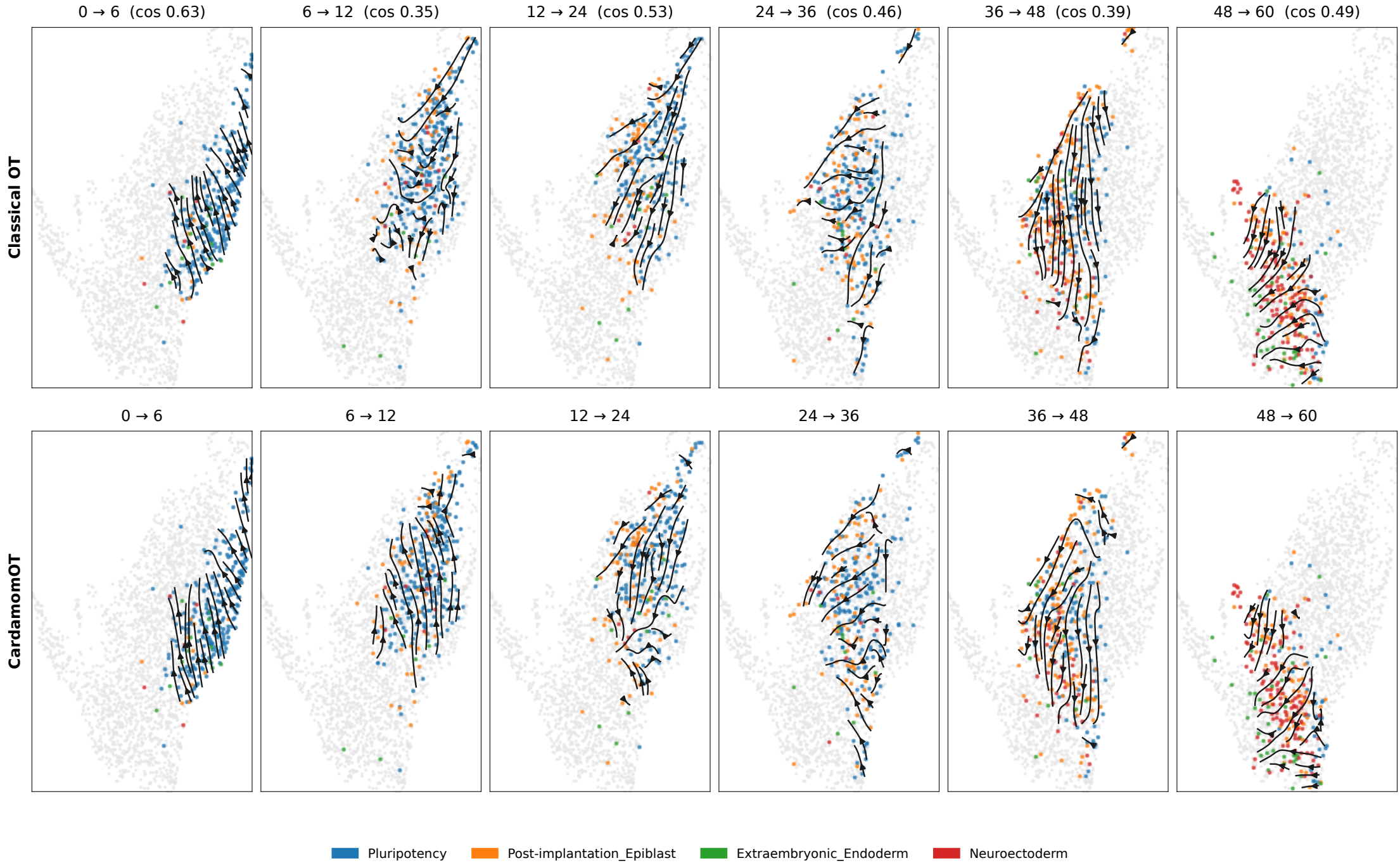
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.53 (PCA space). CardamomOT: final soft couplings of the trajectories (1714 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



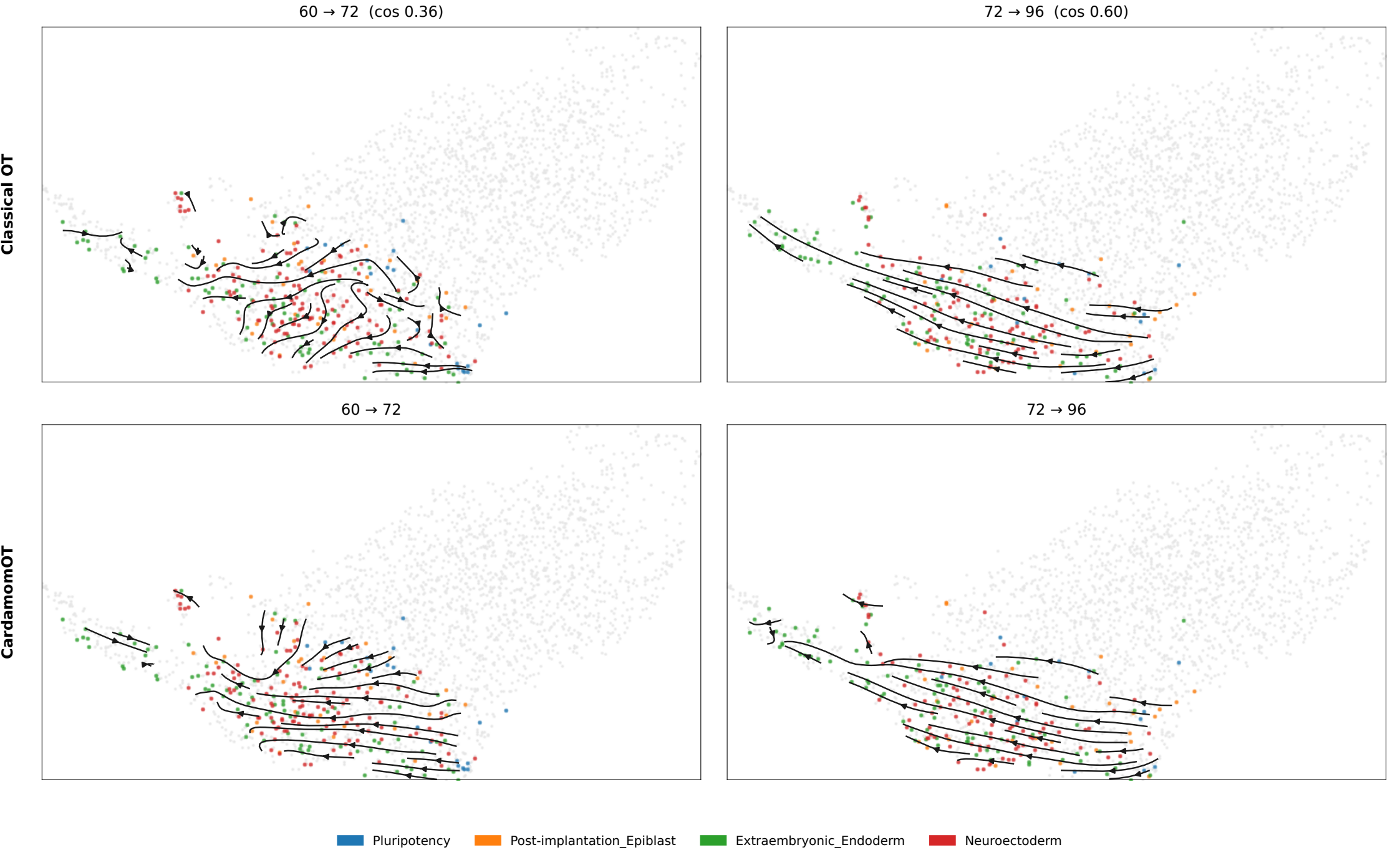
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



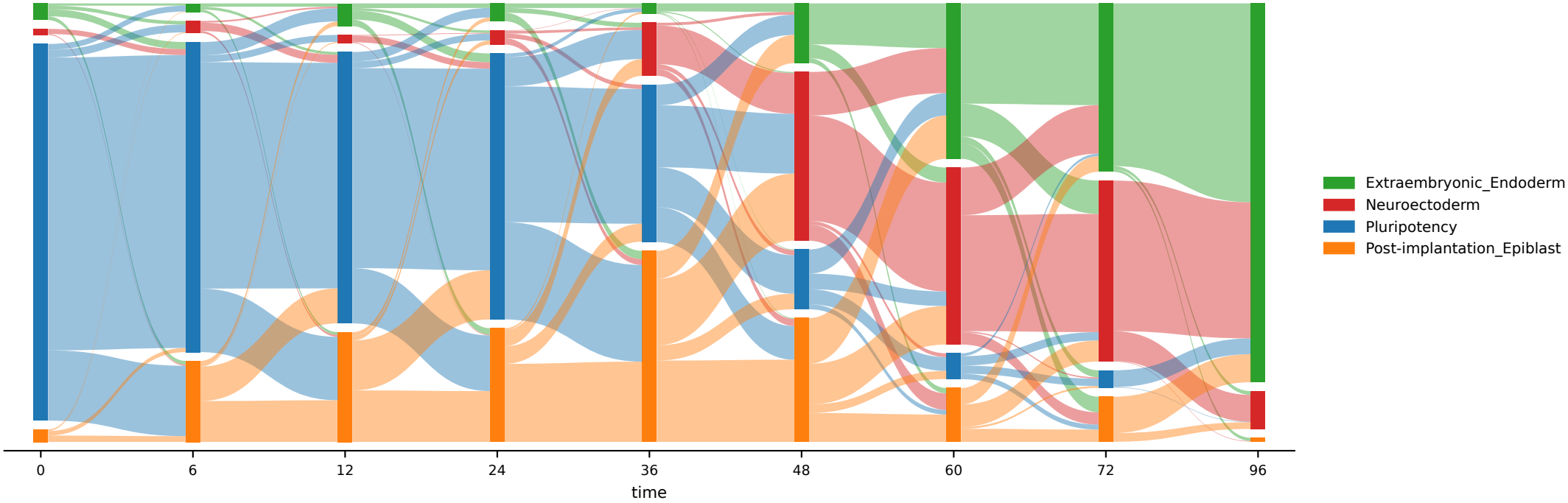
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).

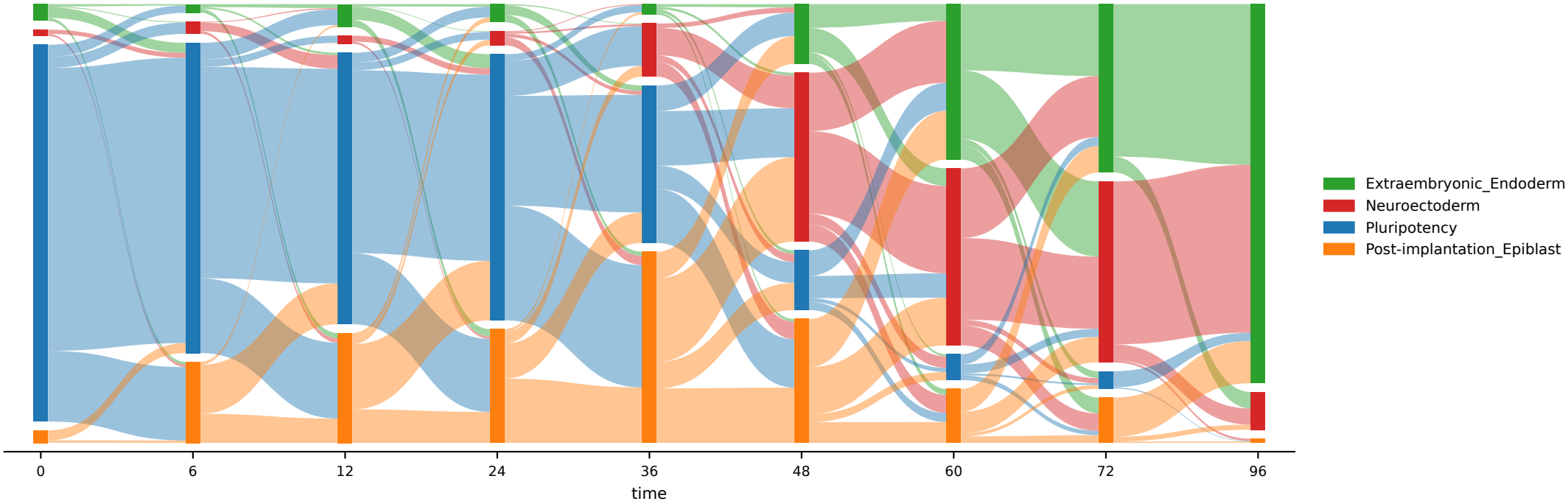


	Top-10 genes up in the ancestors (vs ancestors of the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	**Dab2**	Cd24a	Cdh2	**Krt8**	Meis1	**Gata6**	Hoxb2
Neuroectoderm	**Hoxb2**	**Hoxa1**	**Sox1**	**Sox11**	**Cd24a**	**Pax6**	**Cdh2**	**Meis1**	Dnmt3a	**Nes**
Pluripotency	**Zfp42**	**Esrrb**	**Pou5f1**	**Klf4**	**Klf2**	**Fgf4**	**Sox2**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	Zfp42	**Dppa4**	**Zic3**	**Dnmt3b**	**Dnmt3a**	**Zic2**	**Jarid2**	Klf2	**Pou3f1**

	Top-10 genes up in the cell type (vs the other cells)									
Extraembryonic_Endoderm	**Sparc**	**Lamb1**	**Col4a2**	**Dab2**	**Gata6**	**Krt8**	Pdgfra	Fst	Sox17	Gata4
Neuroectoderm	**Cd24a**	**Hoxb2**	**Hoxa1**	**Sox1**	**Cdh2**	**Sox11**	**Meis1**	**Pax6**	**Nes**	Sparc
Pluripotency	**Zfp42**	**Pou5f1**	**Esrrb**	**Klf2**	**Fgf4**	**Sox2**	**Klf4**	**Jarid2**	**Nr0b1**	**Dppa3**
Post-implantation_Epiblast	**Dppa2**	**Dnmt3a**	**Dppa4**	**Dnmt3b**	**Zic2**	**Zic3**	**Pou3f1**	Wnt8a	Fgf5	**Jarid2**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Extraembryonic_Endoderm	**Col4a2** Cdh2 **Lamb1** Cd24a **Sparc** **Dab2** Meis1 Sox11 Hoxb2 Pax6	**Sparc** **Lamb1** **Col4a2** **Dab2** Gata6 Krt8 Pdgfra Sox17 Fst Gata4
Neuroectoderm	**Hoxb2** **Hoxa1** **Sox1** Lamb1 Gata6 Pou3f1 **Meis1** **Sox11** **Pax6** Pdgfra	**Hoxb2** Cd24a **Hoxa1** **Sox1** Cdh2 **Sox11** **Meis1** **Pax6** Nes Sox9
Pluripotency	**Zfp42** **Esrrb** **Klf2** **Fgf4** **Klf4** **Pou5f1** **Sox2** **Nr0b1** **Jarid2** **Dppa3**	**Pou5f1** **Zfp42** **Esrrb** **Klf2** **Fgf4** **Klf4** **Sox2** **Jarid2** **Nr0b1** **Dppa3**
Post-implantation_Epiblast	Pou5f1 Zfp42 **Dppa2** **Zic3** **Dppa4** **Jarid2** **Dnmt3b** Esrrb Klf4 Klf2	**Dppa2** **Dppa4** Dnmt3a **Dnmt3b** Zic2 Pou3f1 **Zic3** Wnt8a Fgf5 **Jarid2**

Genes of the model (41)

Data/data_train.h5ad, alphabetical order.

Ascl1
Cd24a
Cdh2
Cer1
Col4a2
Dab2
Dnmt3a
Dnmt3b
Dppa2
Dppa3
Dppa4
Esrrb
Fgf4
Fgf5
Fst
Gata4
Gata6
Hoxa1
Hoxb2
Jarid2
Klf2
Klf4
Krt8
Lamb1
Meis1
Nes
Nr0b1
Pax6
Pdgfra
Pou3f1
Pou5f1
Sox1
Sox11
Sox17
Sox2
Sox9
Sparc
Wnt8a
Zfp42
Zic2
Zic3

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) (depth model 'mean') when the depth factor s_i is used.

Depth

not estimated

Reason

41 genes < 10000

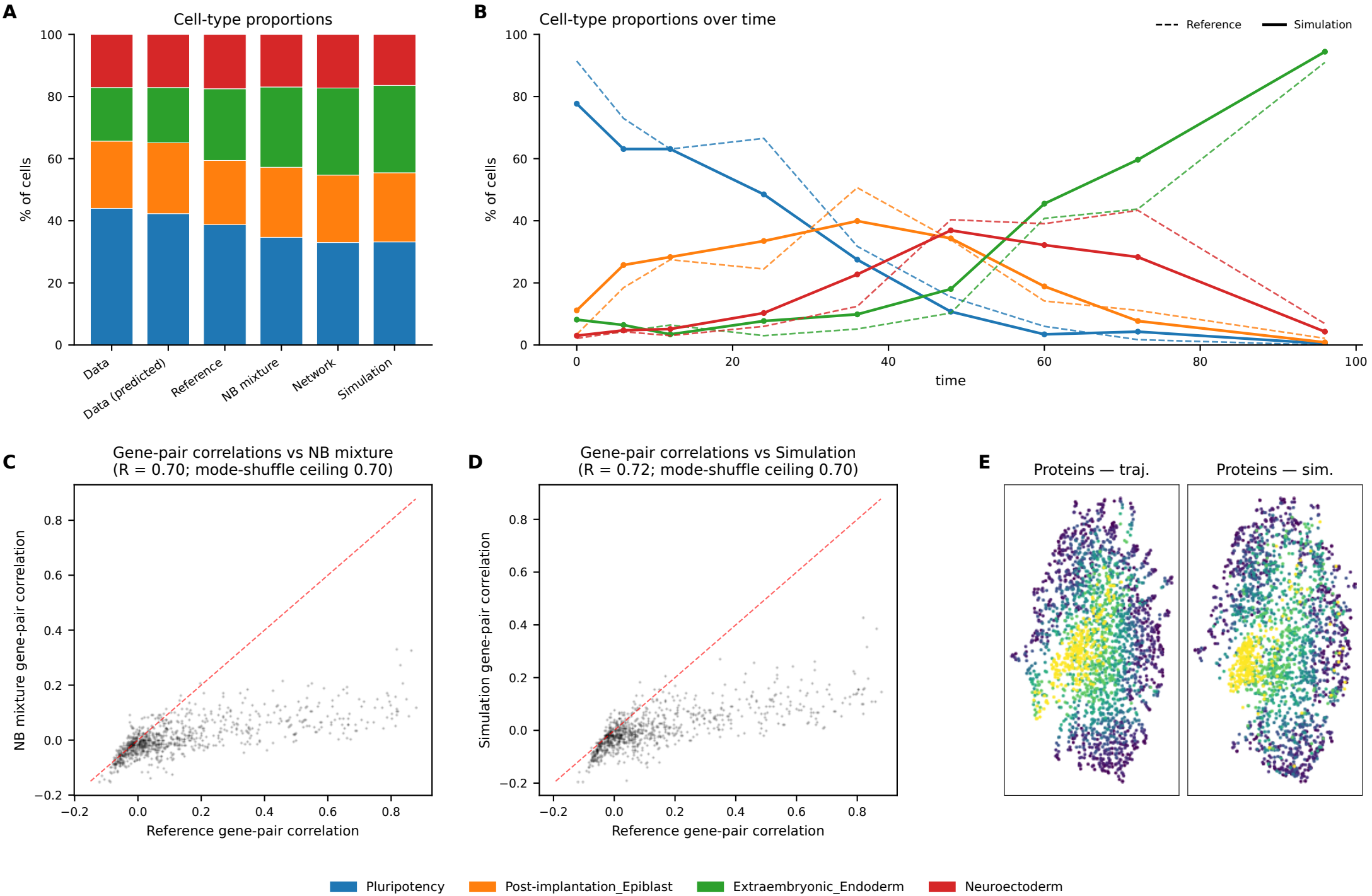
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: states drawn at each time among the observed cells), the NB mixture and network-driven modes along them and the full simulation (without proliferation) are projected onto it.



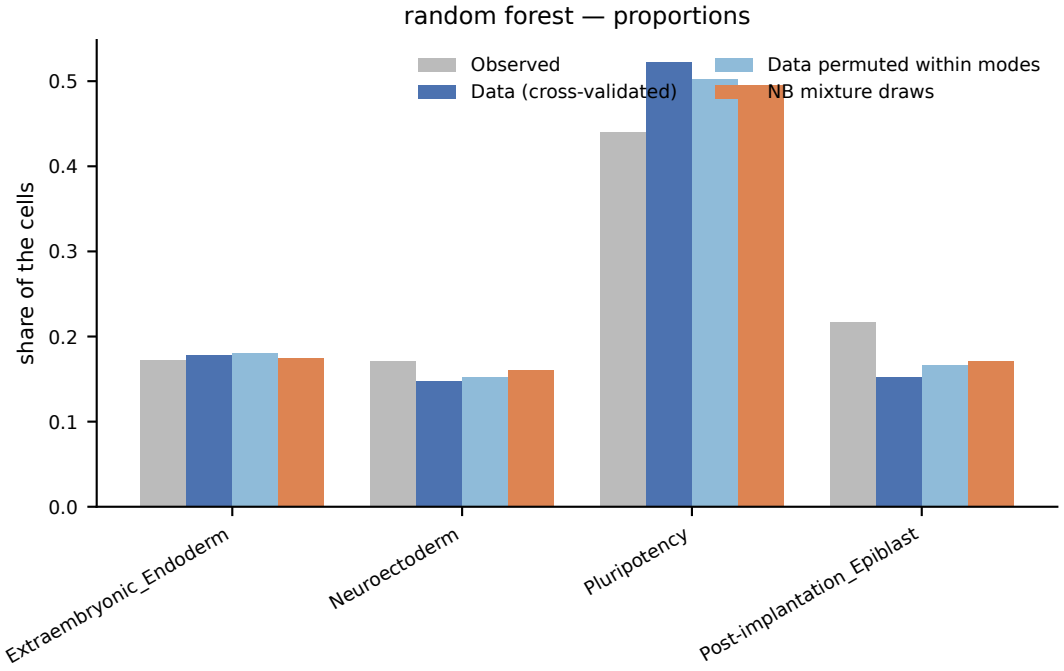
1. Generative model — quantitative checks

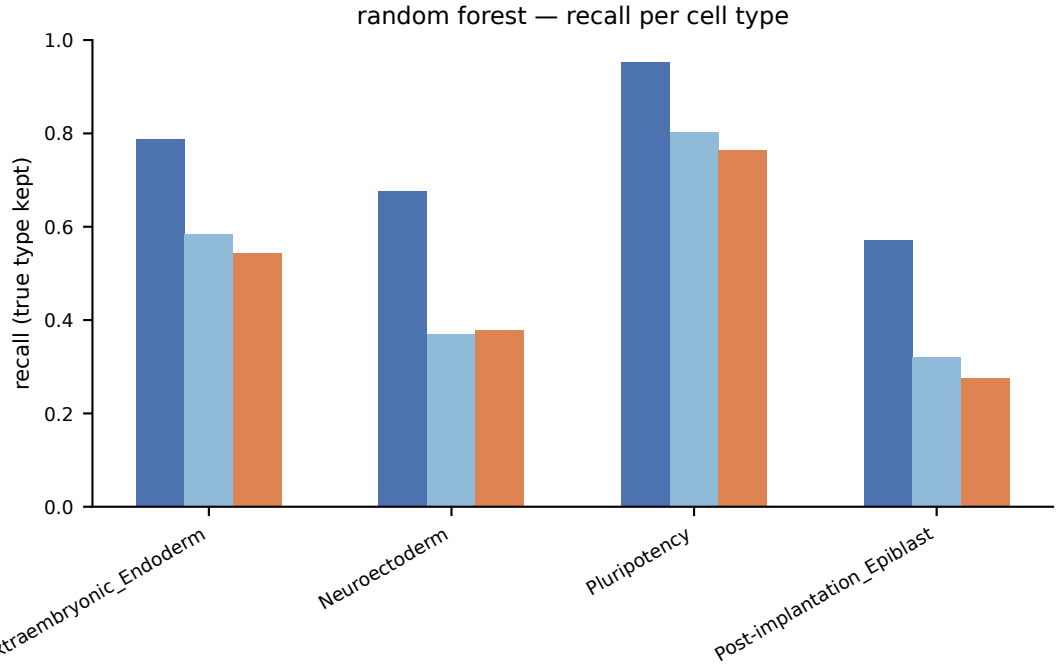
Reference = RNA of the trajectories (states drawn at each time among the observed cells: the observed composition); simulation without proliferation.

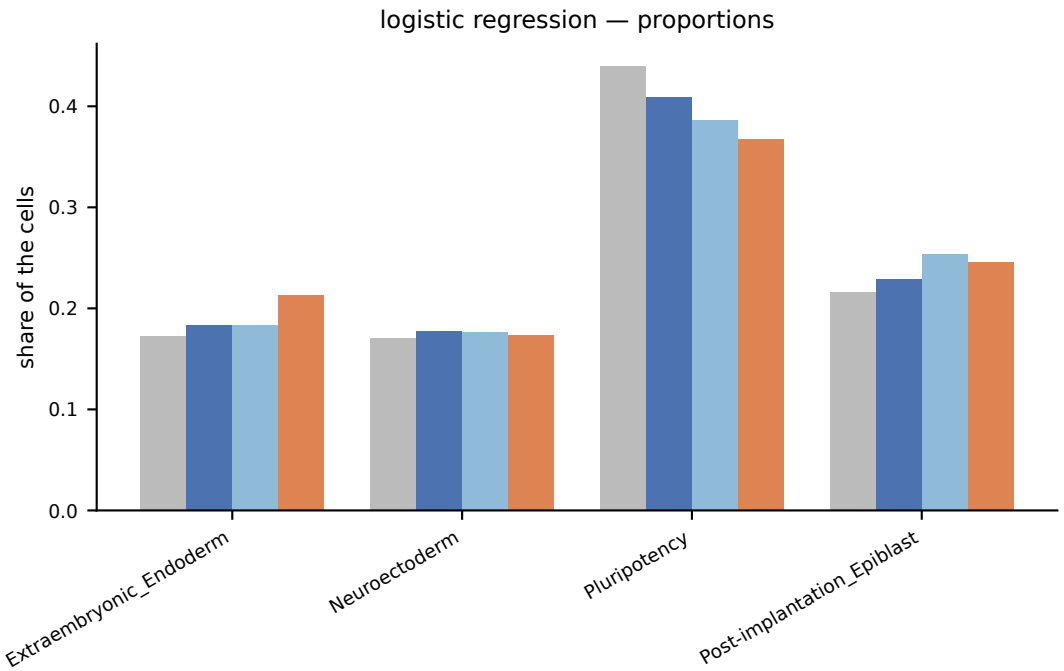


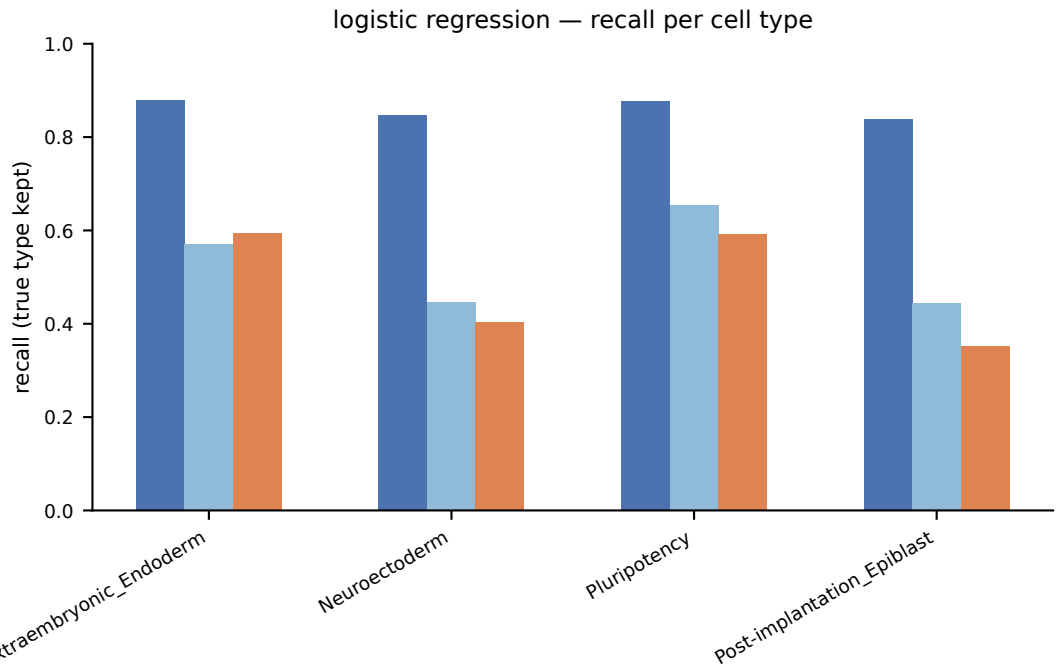
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







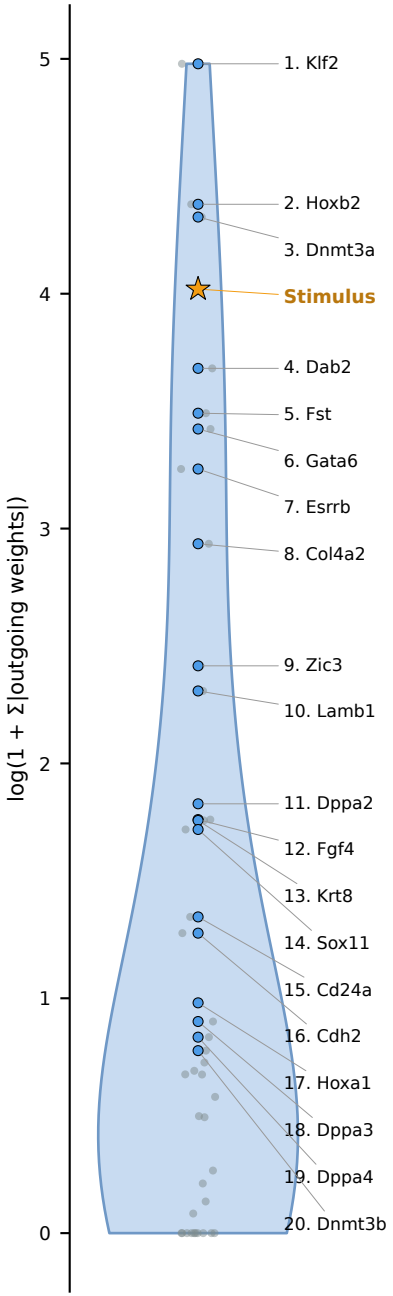


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

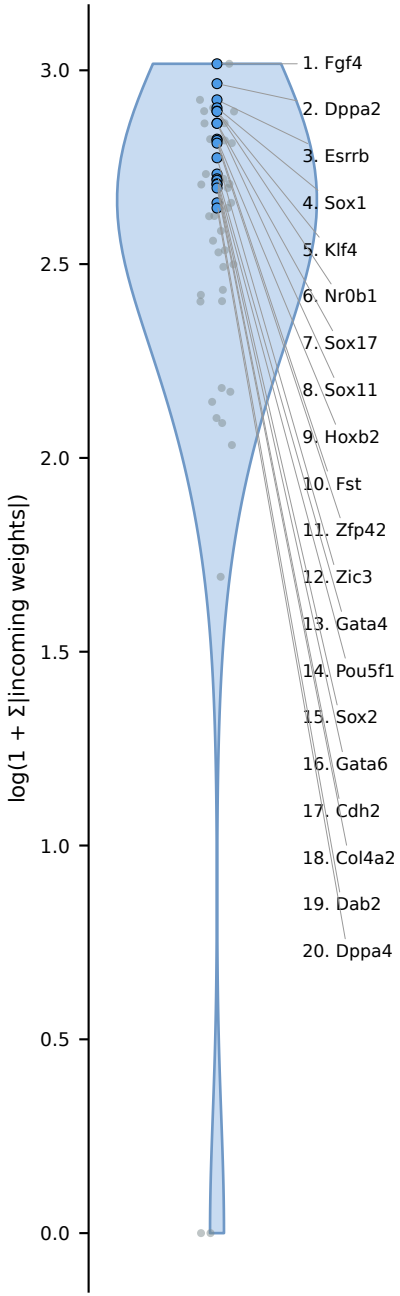
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C Top 20 regulators

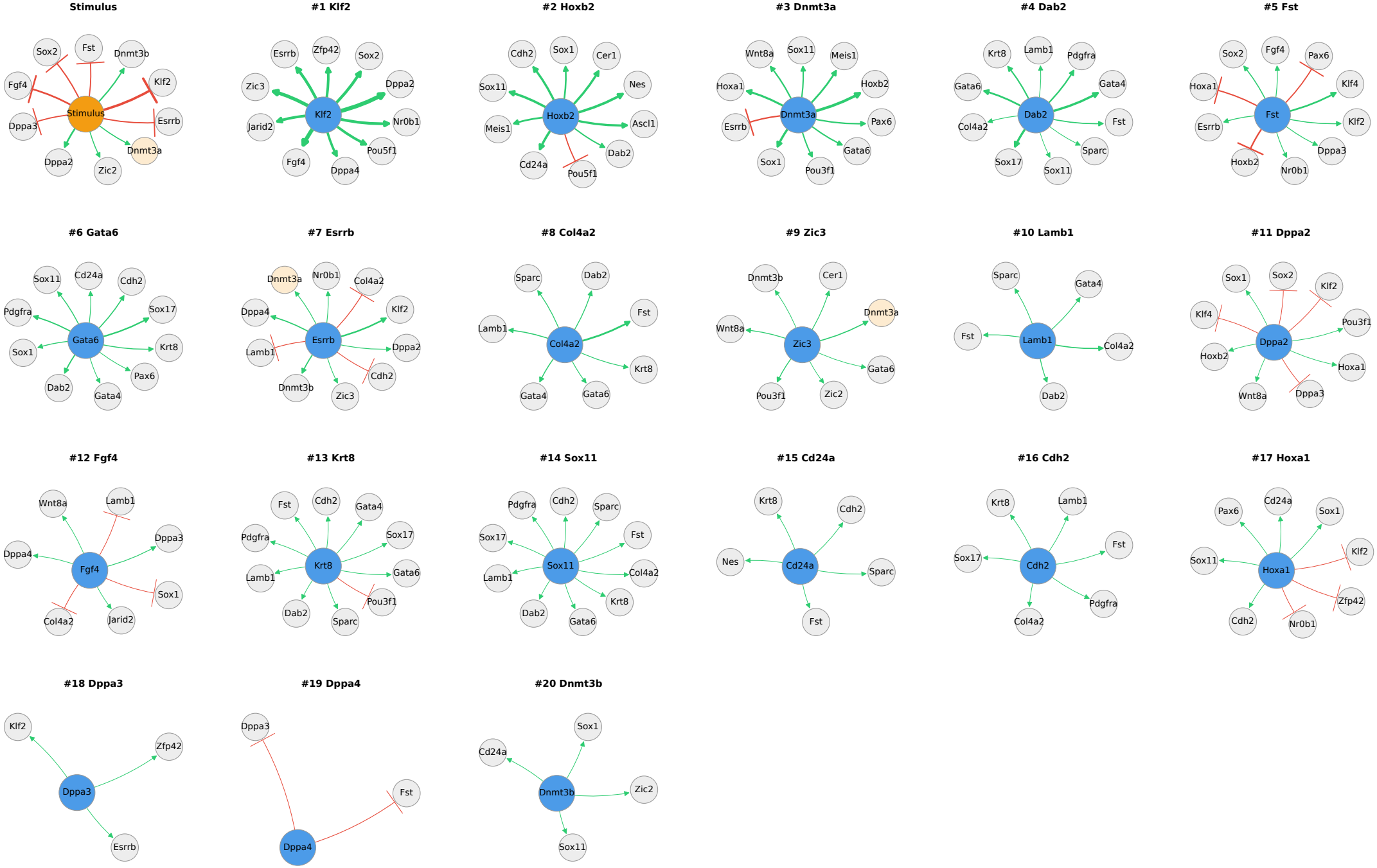
#	Regulator	Σ|w|	act.	inh.
1	Klf2	144.34	13	2
2	Hoxb2	78.90	16	4
3	Dnmt3a	74.68	14	8
4	Dab2	38.73	10	2
5	Fst	31.83	10	3
6	Gata6	29.69	15	3
7	Esrrb	24.88	11	5
8	Col4a2	17.83	7	0
9	Zic3	10.20	7	0
10	Lamb1	9.06	5	0
11	Dppa2	5.22	5	4
12	Fgf4	4.82	4	3
13	Krt8	4.80	9	1
14	Sox11	4.58	10	0
15	Cd24a	2.84	5	0
16	Cdh2	2.59	6	0
17	Hoxa1	1.67	5	3
18	Dppa3	1.46	3	0
19	Dppa4	1.30	0	2
20	Dnmt3b	1.18	4	0
★	Stimulus	54.69	12	7

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Fgf4	19.43	3	1
2	Dppa2	18.41	2	0
3	Esrrb	17.61	3	1
4	Sox1	17.23	6	1
5	Klf4	17.08	2	2
6	Nr0b1	17.06	4	2
7	Sox17	16.52	7	0
8	Sox11	16.51	6	0
9	Hoxb2	15.82	2	1
10	Fst	15.76	9	1
11	Zfp42	15.64	5	3
12	Zic3	15.03	2	0
13	Gata4	14.37	6	0
14	Pou5f1	14.18	2	2
15	Sox2	14.12	2	1
16	Gata6	13.99	7	1
17	Cdh2	13.96	7	1
18	Col4a2	13.82	10	3
19	Dab2	13.27	7	0
20	Dppa4	13.08	4	0

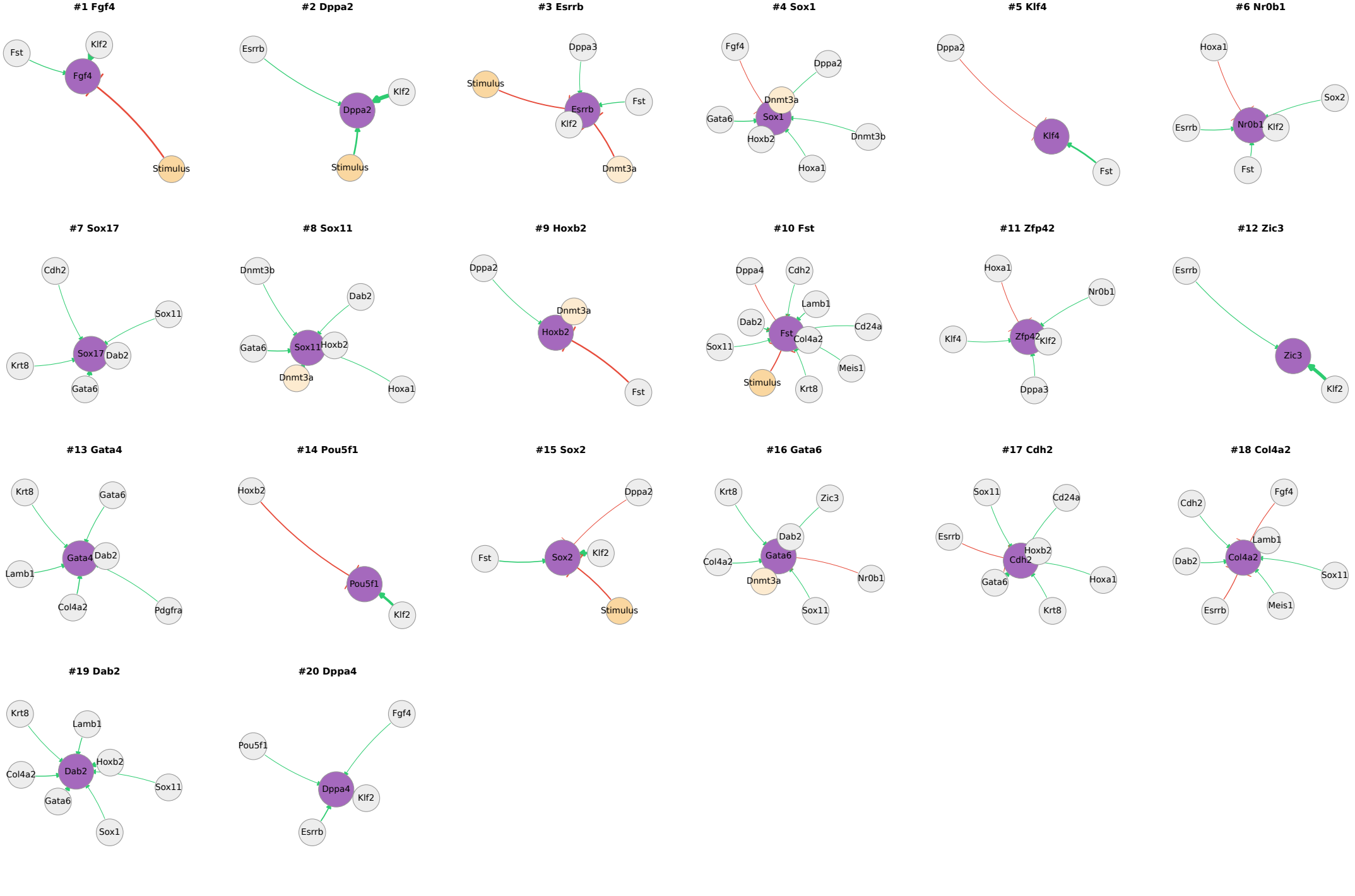
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



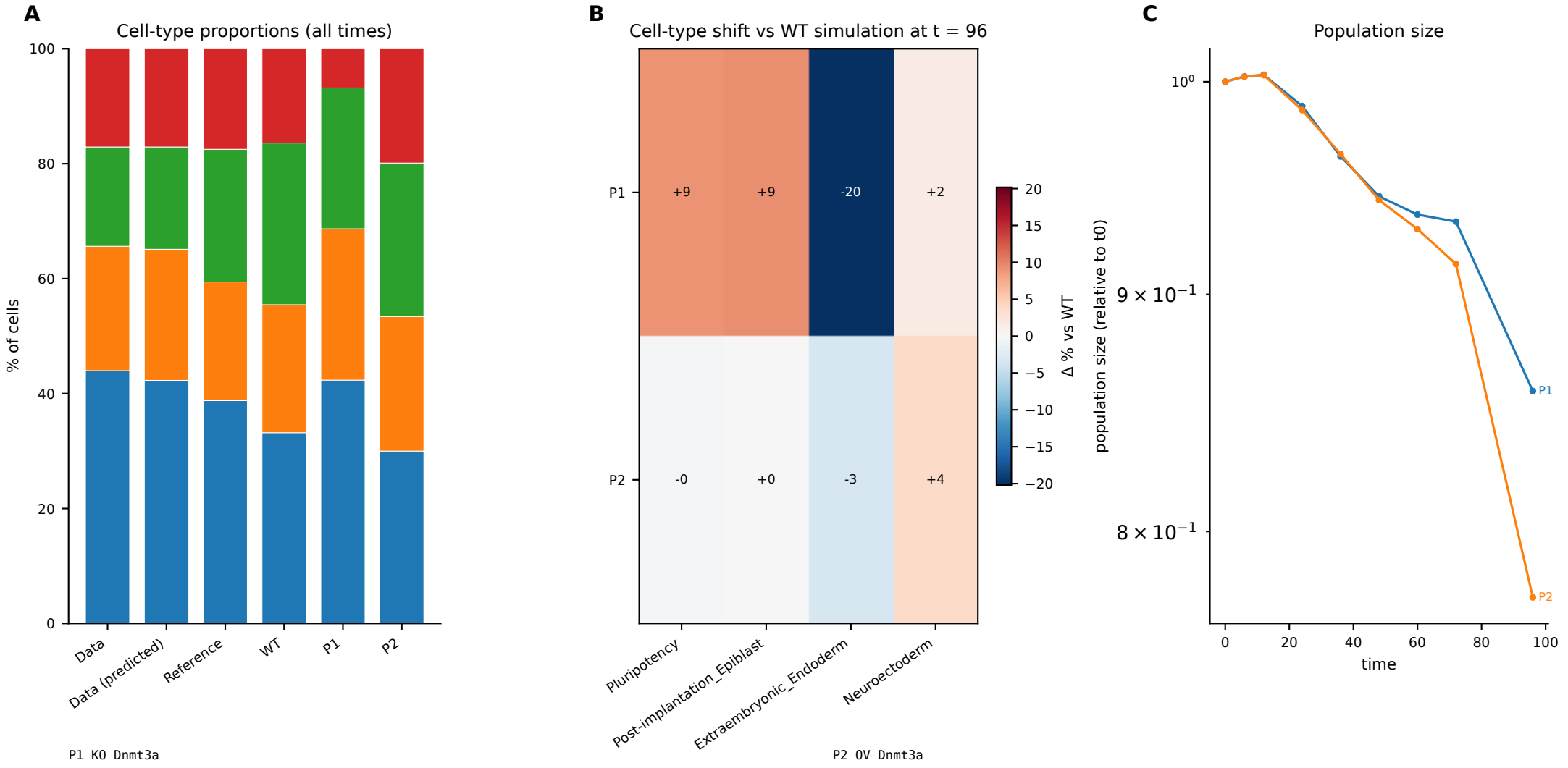
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

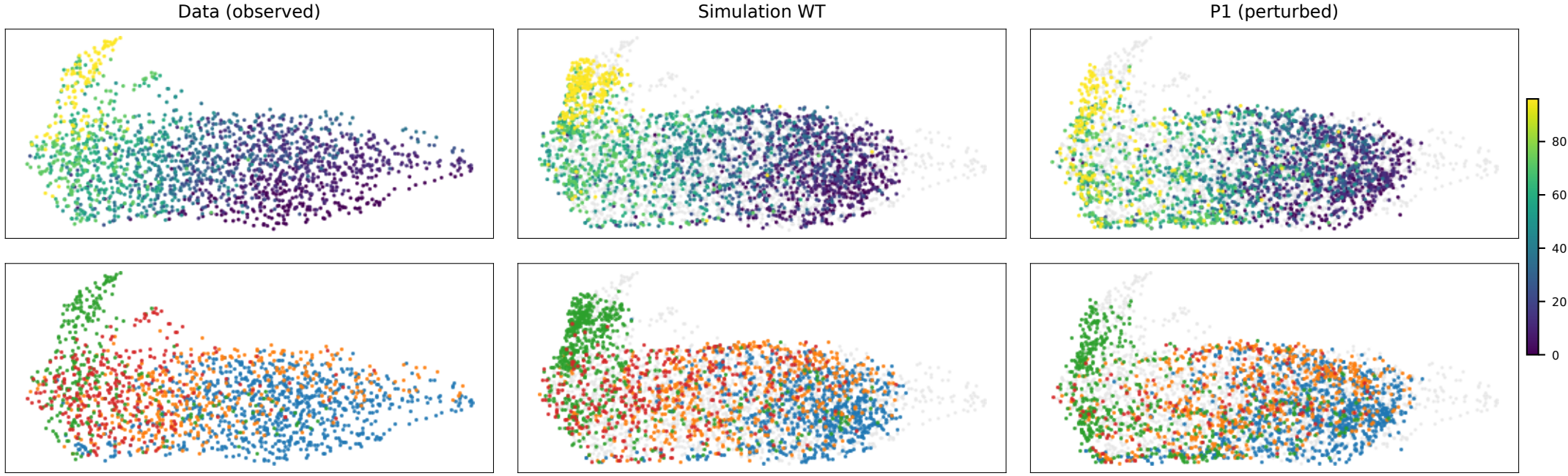
Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (net rates + RATE effects).

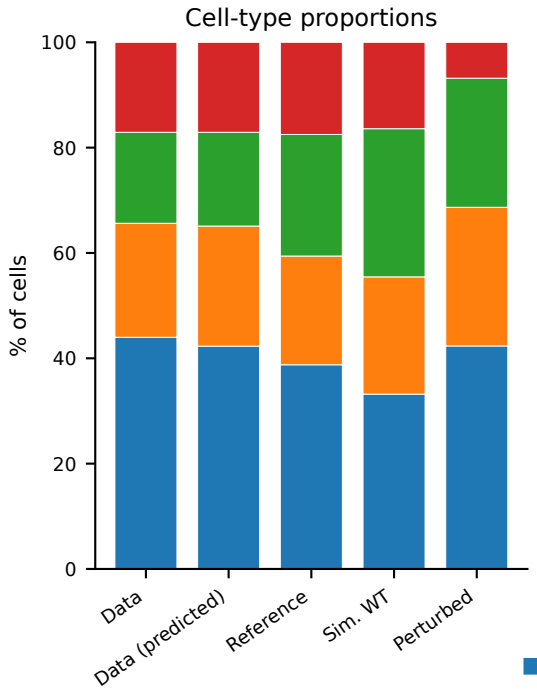


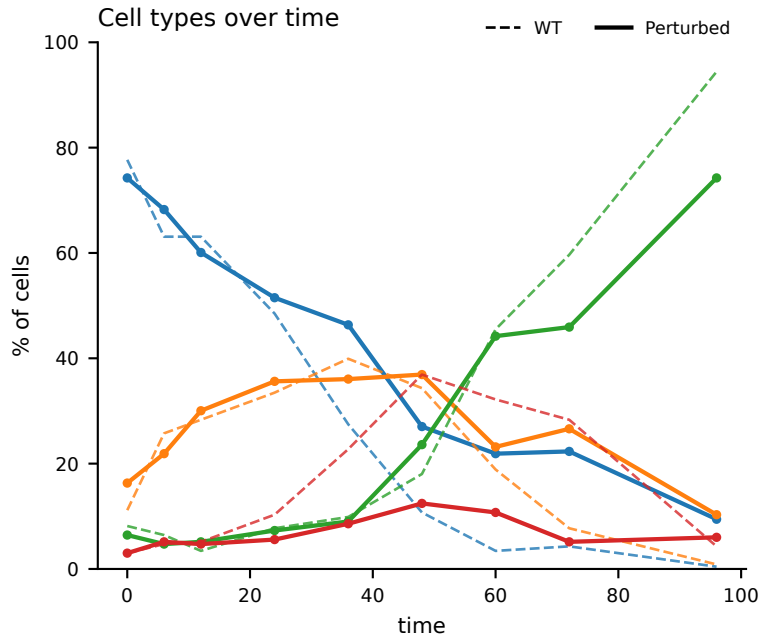
Pluripotency · Post-implantation_Epiblast · Extraembryonic_Endoderm · Neuroectoderm

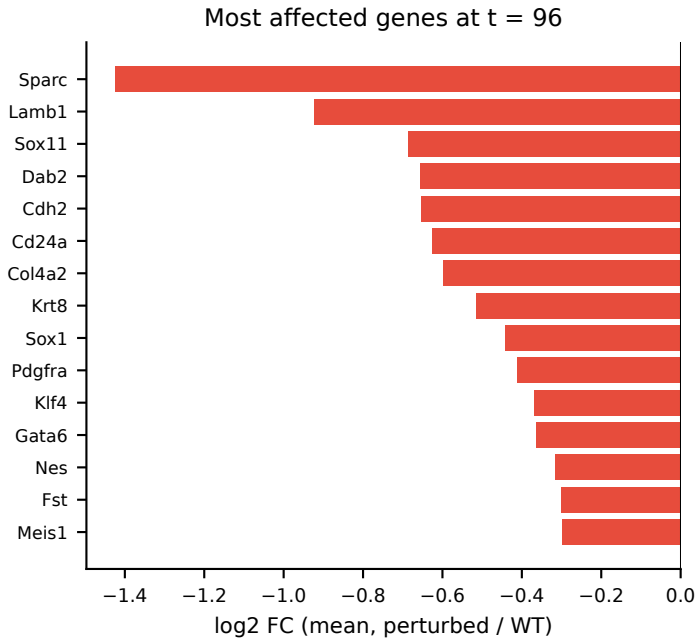
3. Perturbation P1 — KO Dnmt3a

KO_Dnmt3a_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).





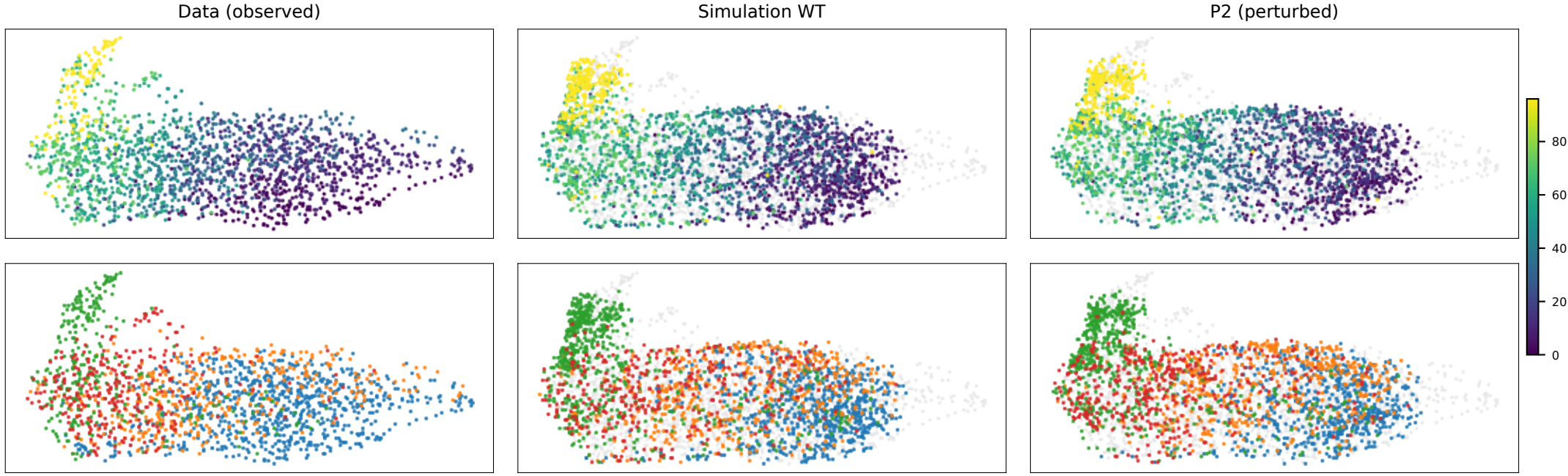


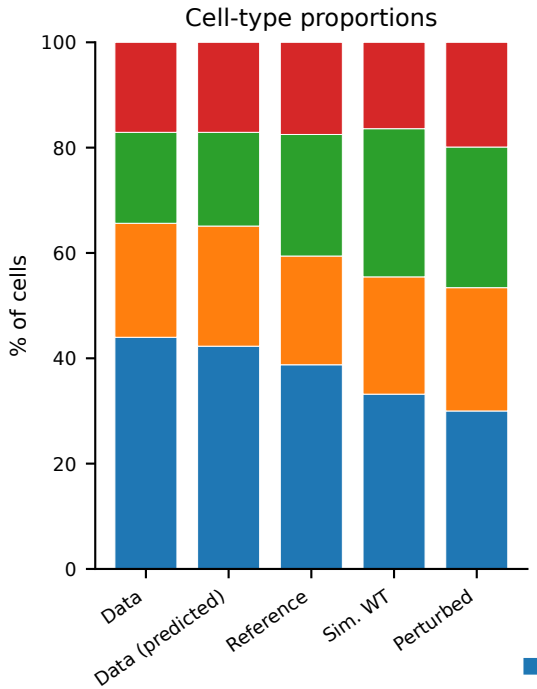


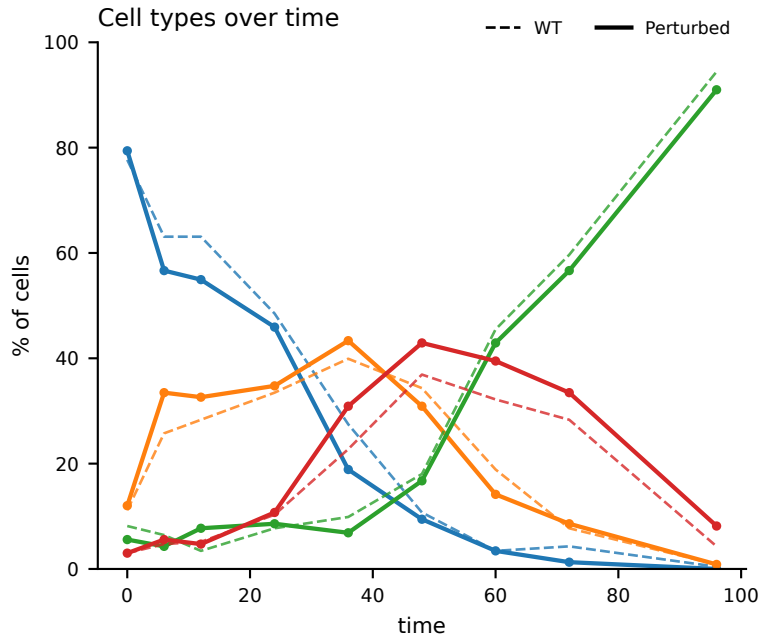


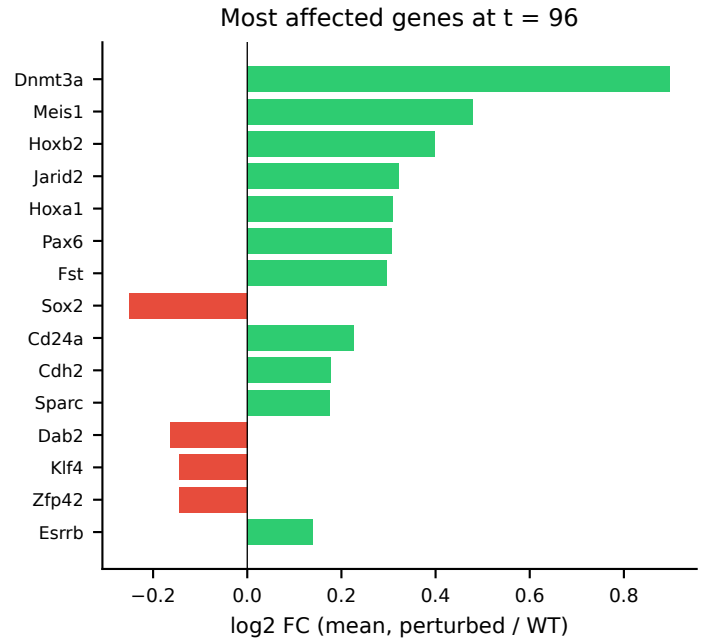
3. Perturbation P2 — OV Dnmt3a

KO_none_OV_Dnmt3a. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).











4. Proliferation

Section unavailable:

```
FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
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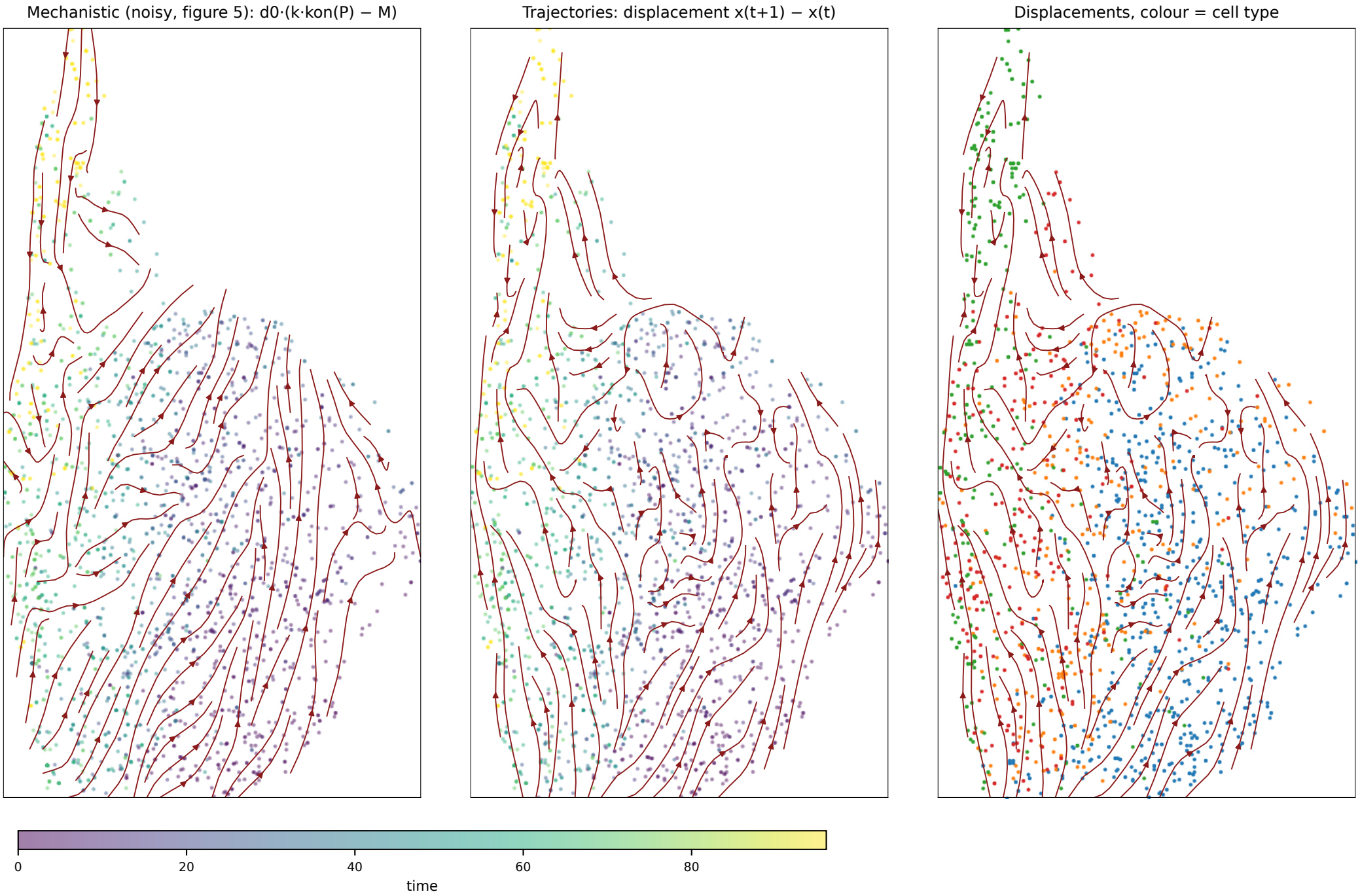
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

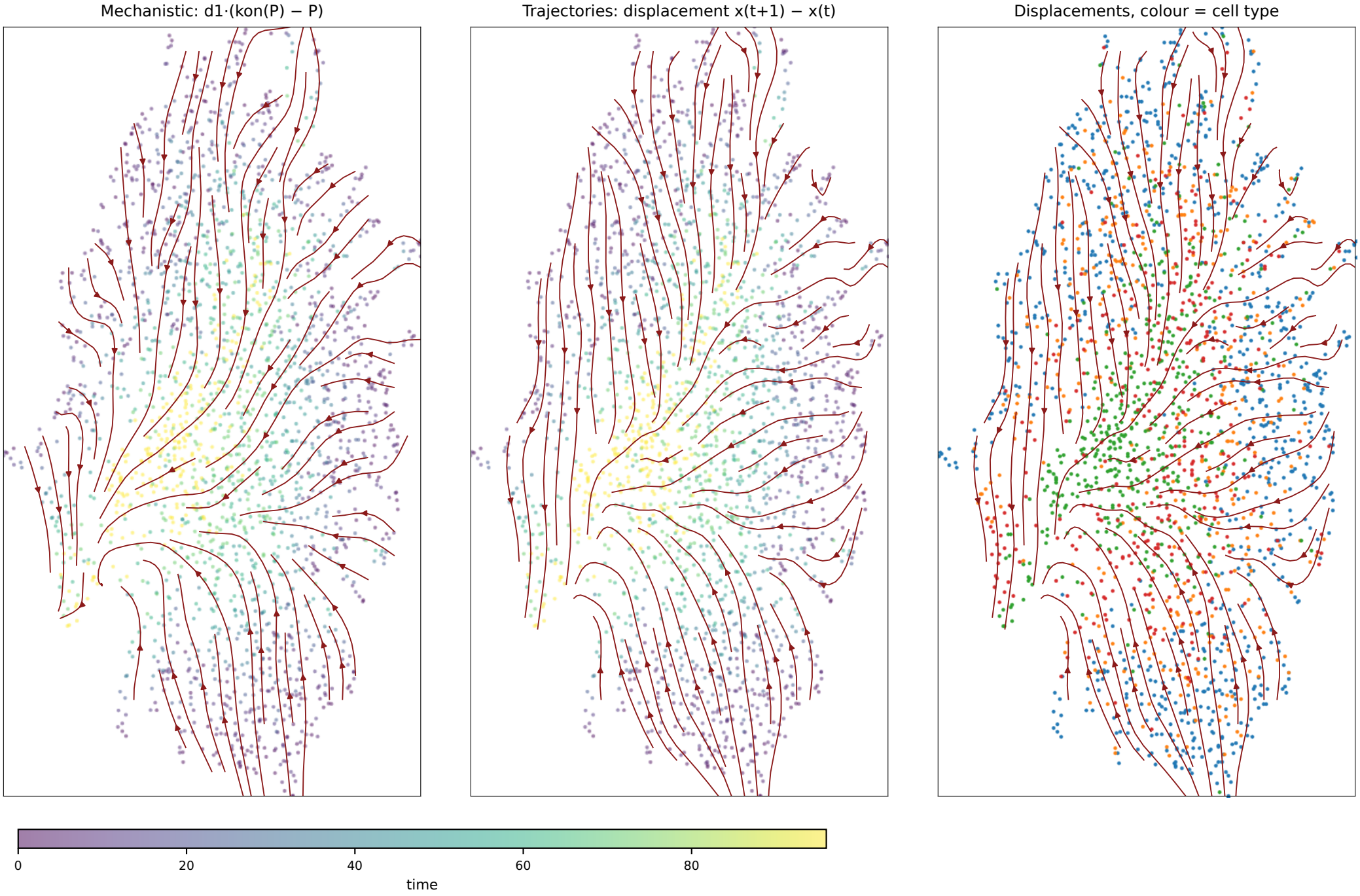
5. Learned dynamics — mRNA velocity and displacement fields

All 2097 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.09.



5. Learned dynamics — Proteins velocity and displacement fields

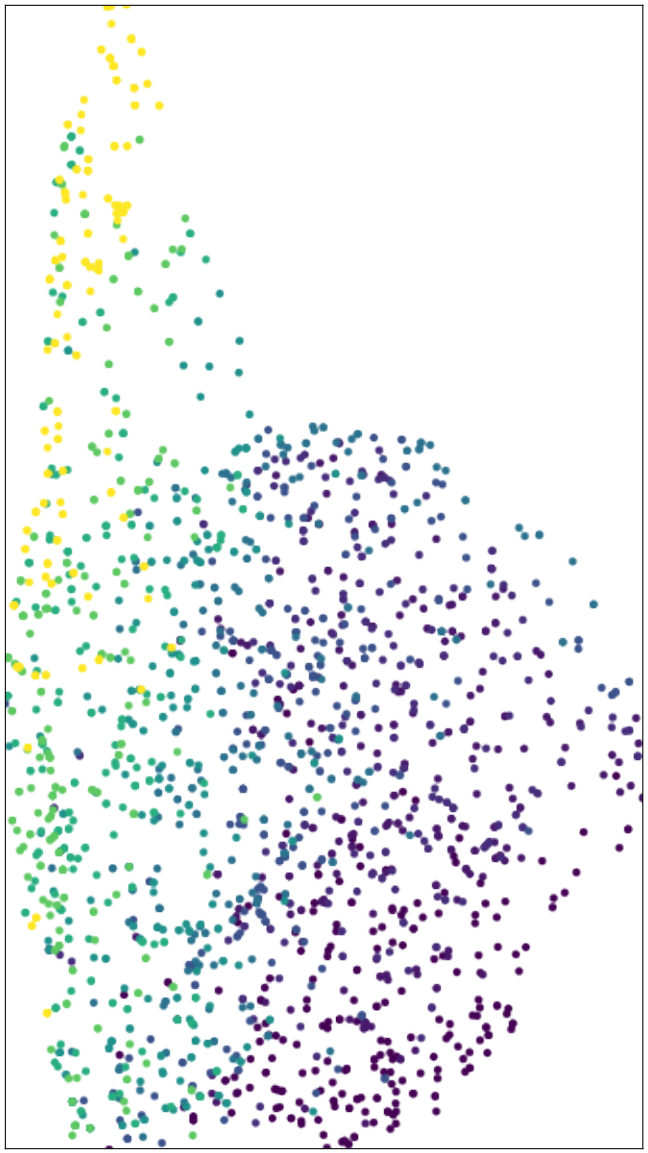
All 2097 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.39.

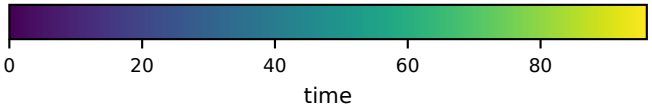


5. Learned dynamics — summary

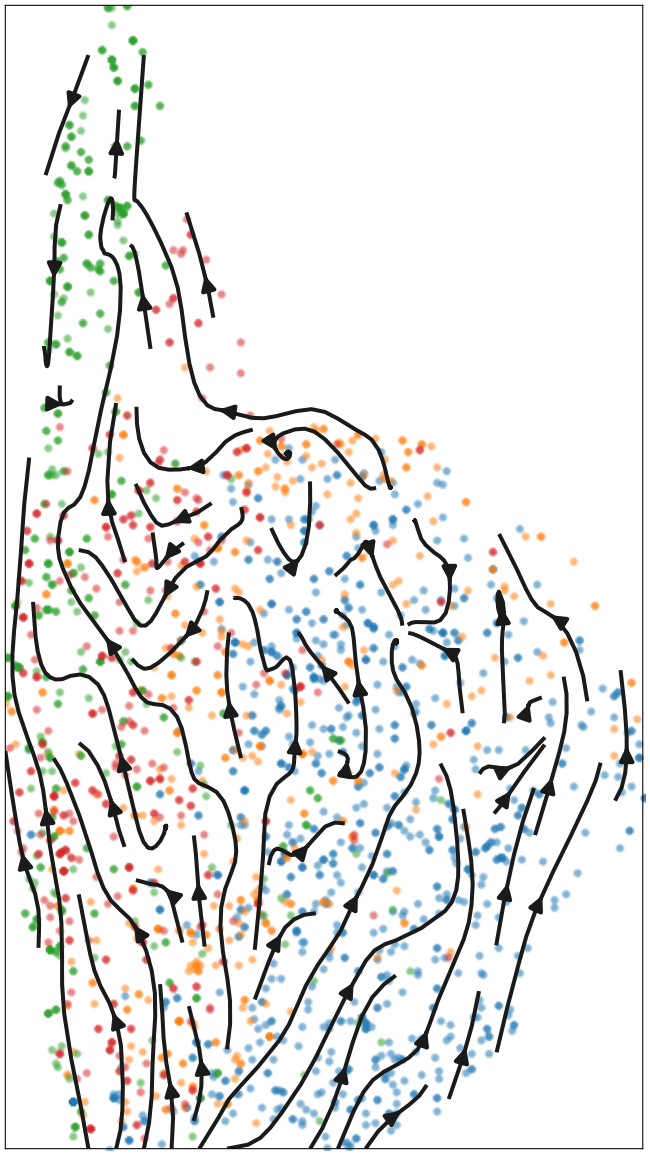
All 2097 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.

Time





mRNA displacement along trajectories, cell_type



No net proliferation rate

Pluripotency
Post-implantation_Epiblast
Extraembryonic_Endoderm
Neuroectoderm

6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

